

Practical no. 23

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include <stdio.h>
#include <conio.h>
#include <stdlib.h>
#define SIZE 5
int queue[SIZE];
int front = -1, rear = -1;
void insert(int value) {
    if ((front == 0 && rear == SIZE - 1) || (rear + 1) % SIZE == front) {
        printf("\nQueue Overflow! (Full)\n");
        return;
    }
    if (front == -1) {
        front = rear = 0;
        queue[rear] = value;
    }
    else {
        rear = (rear + 1) % SIZE;
        queue[rear] = value;
    }
    printf("\n%d inserted into circular queue.\n", value);
}
void delete() {
    if (front == -1) {
        printf("\nQueue Underflow! (Empty)\n");
        return;
    }
    printf("\n%d deleted from circular queue.\n", queue[front]);
    if (front == rear) {
        front = rear = -1;
    }
    else {
        front = (front + 1) % SIZE;
    }
}
void display() {
    int i = front;
    if (front == -1) {
        printf("\nQueue is Empty!\n");
    }
```

```

return;
}
printf("\nCircular Queue elements: ");
while (1) {
printf("%d ", queue[i]);
if (i == rear) break;
i = (i + 1) % SIZE;
}
printf("\n");
}

void main() {
int choice, value;
while (1) {
printf("\n--- Circular Queue Menu ---\n");
printf("1. Insert (Enqueue)\n");
printf("2. Delete (Dequeue)\n");
printf("3. Display\n");
printf("4. Exit\n");
printf("\nEnter choice: ");
scanf("%d", &choice);
switch (choice) {
case 1:
printf("\nEnter value to insert: ");
scanf("%d", &value);
insert(value);
break;
case 2:
delete();
break;
case 3:
display();
break;
case 4:
exit(0);
default:
printf("\nInvalid choice!\n");
}
}
getch();
}

```

OUTPUT :-

```
--- Circular Queue Menu ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit

Enter choice: 1

Enter value to insert: 10

10 inserted into circular queue.
```

```
--- Circular Queue Menu ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit

Enter choice: 2

10 deleted from circular queue.
```

```
--- Circular Queue Menu ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit

Enter choice: 3

Circular Queue elements: 20 30 40 50 60
```